

Algebra 1
Unit 4
Lesson 11 Practice

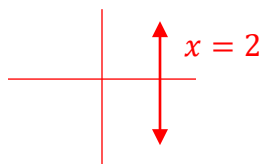
Name Answer Key
Date _____
Period _____

1. Write the equation of the line that passes through $(4, 2)$ and is perpendicular to $y = 3x + 1$ in point-slope form.

$$m = -\frac{1}{3} \quad y - 2 = -\frac{1}{3}(x - 4)$$

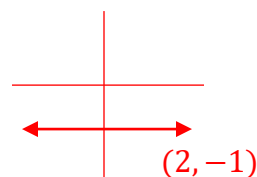
2. Write the equation in slope-intercept form of the line that is perpendicular to $x = 2$ and passes through point $(2, -1)$.

Vertical line



Horizontal line

$$y = -1$$



3. Write the equation of the line that contains point $(-5, 5)$ and is perpendicular to $\frac{5}{9}x - y = 4$ in standard form.

$$y = \frac{5}{9}x - 4$$

$$\perp m = -\frac{9}{5}$$

$$y - 5 = -\frac{9}{5}(x + 5)$$

$$y - 5 = -\frac{9}{5}x - 9$$

$$+9 \quad +9$$

$$y + 4 = -\frac{9}{5}x$$

$$(5)y + \frac{9(5)}{5}x = -4(5)$$

$$9x + 5y = -20$$

4. Write the equation of the line that is perpendicular to $2x + 5y = 6$ and passes through the origin. $(0, 0)$

$$\perp m = \frac{5}{2} \quad (0, 0)$$

$$y = \frac{5}{2}x$$

$$\frac{5y}{5} = \frac{6-2x}{5}$$

$$y = -\frac{2}{5}x + \frac{6}{5}$$

5. Line F passes through $(-12, 0)$ and is perpendicular to Line Z which has an undefined slope. What is the slope of line F ?

$x = \text{line}$

Slope of line F is 0 since it is a horizontal line.

6. Write the equation of the line that is perpendicular to the line that passes through the points $(7, -3)$ and $(7, 14)$ and passes through point $(4, -\frac{1}{2})$.

x	y
7	-3
7	14

$$0 < 7 \quad 14 > -3$$

$$m = \frac{17}{0}$$

$$y = -\frac{1}{2}$$

Undefined $\rightarrow$ vertical line – perpendicular is a horizontal line.

7. Write the equation of a line that is perpendicular to $\frac{-y}{-1} = \frac{-x}{-1}$ and contains point $(-1, -1)$.

Slope of $\perp$ line = -1

$$y = x \quad m = 1$$

$$y + 1 = -(x + 1) \text{ or } y = -x -$$

8. Write the equation of the line, in all three forms, that passes through point $(6, -2)$ and is perpendicular to $y + 3 = -7(x + 1)$. $m = -7 \perp \text{slope} = \frac{1}{7}$

Point-Slope Form	Slope-Intercept Form	Standard Form
$y + 2 = \frac{1}{7}(x - 6)$	$y + 2 = \frac{1}{7}x - \frac{6}{7}$ $\quad \quad -2 \quad \quad -2$ $y = \frac{1}{7}x - \frac{20}{7}$	$(7)y = \frac{1(7)}{7}x - \frac{20}{7}(7)$ $7y = x - 20$ $\quad \quad -x \quad \quad -x$ $-x + 7y = -20$ or $x - 7y = 20$

9. Write the equation of the line, in all three forms, that is perpendicular to the line that passes through the points $(20, 0)$ and $(0, -10)$ and passes through point $(-10, 10)$.

$$\begin{array}{c|c} x & y \\ \hline 20 & 0 \\ 0 & -10 \end{array} \begin{array}{l} \nearrow \\ \searrow \end{array}$$

$$m = \frac{-10}{-20} = \frac{1}{2}$$

$$\perp m = -2$$

Point-Slope Form	Slope-Intercept Form	Standard Form
$y - 10 = -2(x + 10)$	$y - 10$ $= -2x - 20$ $\quad \quad +10 \quad \quad +10$ $y = -2x - 10$	$2x + y = -10$

10. Write the equation of the line, in all three forms, that is perpendicular to the line that passes through the points $(8, 9)$ and $(-4, -7)$ and passes through point $(0, -5)$.

$$\begin{array}{c|c} x & y \\ \hline 8 & 9 \\ -4 & -7 \end{array} \begin{array}{l} \nearrow \\ \searrow \end{array}$$

Point-Slope Form	Slope-Intercept Form	Standard Form
$y + 5 = -\frac{3}{4}(x - 0)$ OR $y + 5 = -\frac{3}{4}x$	$y = -\frac{3}{4}x - 5$	$(4)y = \frac{-3(4)}{4}x - 5(4)$ $4y = -3x - 20$ $\quad \quad +3x \quad \quad +3x$ $3x + 4y = -20$

$$\frac{\Delta y}{\Delta x} = \frac{-16}{-12} = \frac{8}{6} = \frac{4}{3}$$

$$\text{So } \perp m = -\frac{3}{4}$$